THERMAL PROPERTIES OF MATTER

1. For an ideal gas, coefficient of volume expansion at constant pressure is equal to

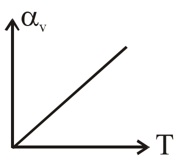
(1) T

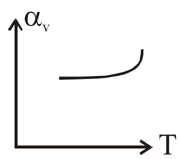
(2) $\frac{1}{T}$

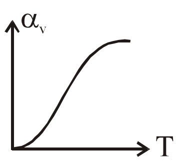
(3) T^2

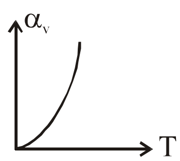
(4) Not depends on temp.

2. Coefficient of volumetric expansion α_v is not a constant. It depends on temperature. Variation of α_v with temperature for metal is.

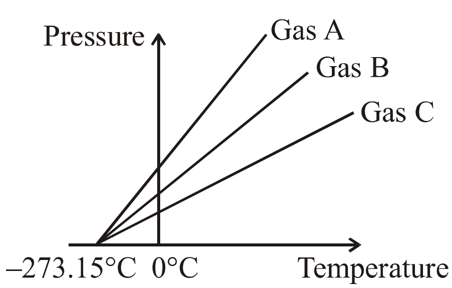
(1) 

(2) 

(3) 

(4) 

3. The absolute minimum temperature for an ideal gas therefore inferred by extrapolating the straight line to the axis, in the figure given below. This temperature is found to be –273.15°C and is designated as



(1) Kelvin temperature (2) Celsius temperature

(3) Low temperature (4) Absolute zero

4. Cooking is difficult on hills because

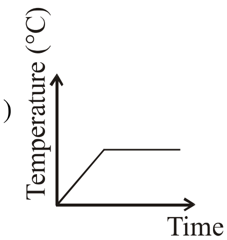
(1) atmospheric pressure is higher

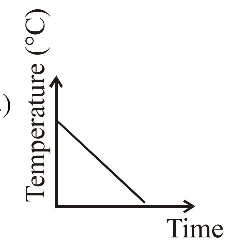
(2) atmospheric pressure is lower

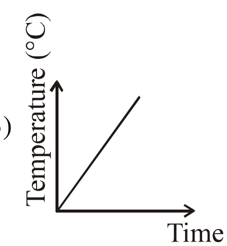
(3) boiling point of water is reduced

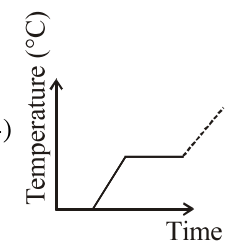
(4) Both (2) and (3)

5. A block of ice at 0°C is slowly heated and converted into steam at 100°C. Which of these curves represents the phenomenon quantitatively?

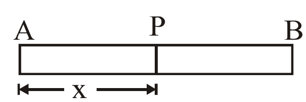
(1) 

(2) 

(3) 

(4) 

6. Heat is flowing steadily from A to B. Temperature T at point P, at distance x from A is such that :-



(1) T decreases linearly with x

(2) T increases linearly with x

(3) T decreases exponentially with x

(4) T increases with x as $T \propto x^2$

7. Heat is associated with :-

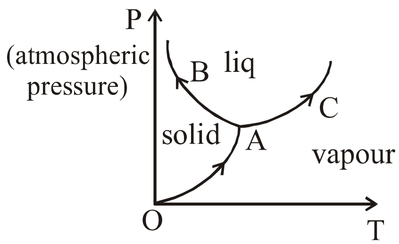
(1) Kinetic energy of random motion of molecules

(2) Kinetic energy of orderly motion of molecules

(3) Total kinetic energy of random and orderly motion of molecules

(4) Kinetic energy of random motion in some cases and kinetic energy of orderly motion in other

8. For the phase diagram of water given in figure, curves OA, AB and AC are respectively.



(1) Sublimation curve, vaporisation curve, and fusion curve

(2) Sublimation curve, fusion curve and vaporisation curve

(3) Fusion curve, vaporisation curve and sublimation curve

(4) Fusion curve, sublimation curve, and vaporisation curve

9. Which of the following statements is/are correct?

(I) Convection is a mode of heat transfer by actual motion of matter.

(II) Convection is possible only in gases.

(III) Convection can be natural or forced.

(1) Only I

(2) Only II

(3) Both I and III

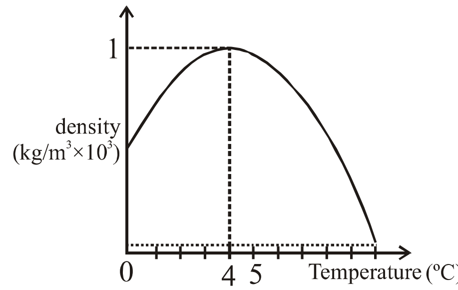
(4) None of these

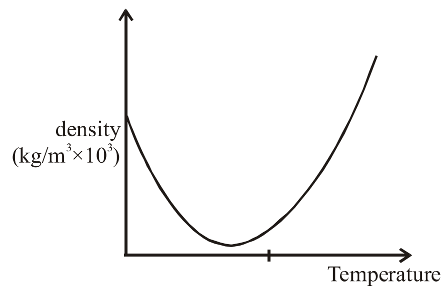
10. Find out the increase in moment of inertia I of a uniform rod (Coefficient of linear expansion α) about its perpendicular bisector when its temperature is slightly increased by ΔT.

(1) $2I\alpha\Delta T$ (2) $4I\alpha\Delta T$

(3) $6I\alpha\Delta T$ (4) $3I\alpha T$

11. Which of the following graph shows the variation of density of water with increase in temperature.

(1)


(2)


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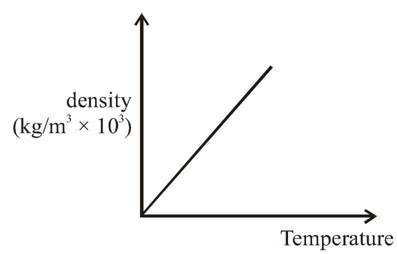
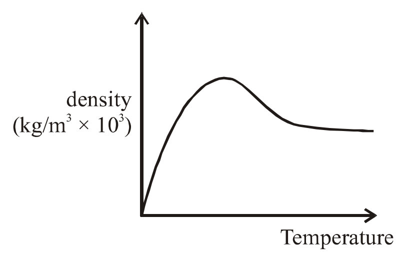


(4)


12. Water is used as coolant in automobiles radiators because.

(1) It has low density

(2) It has low specific heat

(3) It has high specific heat

(4) It has high density

13. Melting of ice at low temperature caused by pressure is

(1) Regelation

(2) Sublimation

(3) Vaporisation

(4) Fusion

14. Which of the following statements are correct.
(a) The triple point of water is 253.16 K
(b) Burns from steam are more severe than these from boiling water
(c) Ethyl alcohol expands less than mercury for the same rise in temperature
(d) When fully inflated balloon is immersed in cold water, it will contract.
(1) a, b, c, d (2) b, c
(3) b, c, d (4) b, d

15. For transmission of heat from one place to the other, medium is required in
(1) Conduction (2) Convection
(3) Radiation (4) (1) and (2) both

16. The equatorial and polar regions of the earth receive unequal heat. The natural convection current arising due to this is called.
(1) Land breeze (2) Sea bneeze
(3) Trade wind (4) Tornado

17. The thermal radiation emitted by body is proportional to T^4, where T is absolute temperature, is valid for.
(1) Ideal black body
(2) All bodies
(3) Bodies painted black only
(4) Polished bodies only

18. Newton's law of cooling is a special case of
(1) Wien's displacement law
(2) Kirchoff's law
(3) Stefan's law
(4) Plank's law

19. The thermal conductivity of rod depends on
(1) Length
(2) Mass
(3) Area of cross section
(4) Material of the rod

20. A bimetallic strip is made of aluminium and steel ($\alpha_{al} > \alpha_{steel}$). On heating the strip will :-
(1) remain straight
(2) get twisted
(3) will bend with aluminium on concave side
(4) will bend with steel on concave side

21. A uniform metallic rod rotates about its perpendicular bisector with constant angular speed. If it is heated uniformly to raise its temperature.
(1) its speed of rotation increases
(2) its speed of rotation decreases
(3) its speed of rotation remain the same
(4) insufficient data

22. A sphere is dipped in water. Which of following is true ?
(1) Buoyancy will be maximum at 4ºC
(2) Buoyancy will be maximum at 0ºC
(3) Buoyancy will be same at 0ºC and 4ºC
(4) Buoyancy will be less at 4ºC

23. As temperature is increased, the time period of pendulum.
(1) increases as its length increases even though its centre of mass still remains at centre of the bob
(2) decreases as its length increases even though its centre of mass still remains at centre of the bob
(3) increases as its length increases due to shifting of centre of mass below centre of the bob
(4) decreases as its effective length remains same but centre of mass shift above centre of bob

24. A sphere, cube and a thin plate, all of same material and mass, are initially heated to same high temperature. Then
(1) Plate will cool fastest and cube the slowest
(2) Sphere will cool fastest and cube the slowest
(3) Plate will cool fastest and sphere the slowest
(4) cube will cool fastest and plate the slowest

25. Choose the correct option.

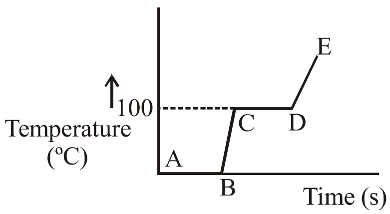
(1) X and Y are in thermal equilibrium but not X and Z. Then Y and Z may be in thermal equilibrium

(2) X and Y are in thermal equilibrium but not X and Z. Then Y and Z must be in thermal equilibrium

(3) X is neither in equilibrium with Y nor with Z. Then Y and Z must be in equilibrium.

(4) X is neither in equilibrium with Y nor with Z. Then Y and Z may be in thermal equilibrium.

26. From the graph shown for ice on heating, select the correct option.



(1) The region AB represents ice and water in equilibrium.

(2) At B water starts boiling

(3) At C all water gets converted to steam

(4) DE region represents phase change from water to steam.

27. Which of the following is used in making a pendulum clock ?

(1) Iron (2) Stainless steel

(3) Invar (4) Graphite

28. A cooking pot is coated black because.

(1) Black substances absorb more heat

(2) Black substances reflect more heat

(3) Black surfaces are easier to clean

(4) The material of the pot would not be damaged

29. A cup of hot tea kept on a metal table in a room loses heat by.

(1) Conduction (2) Convection

(3) Radiation (4) All of above